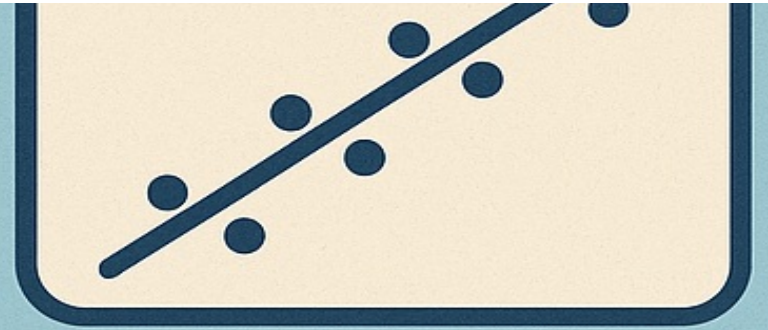
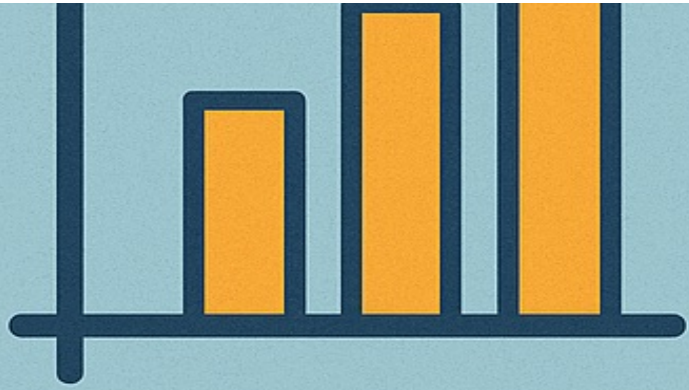
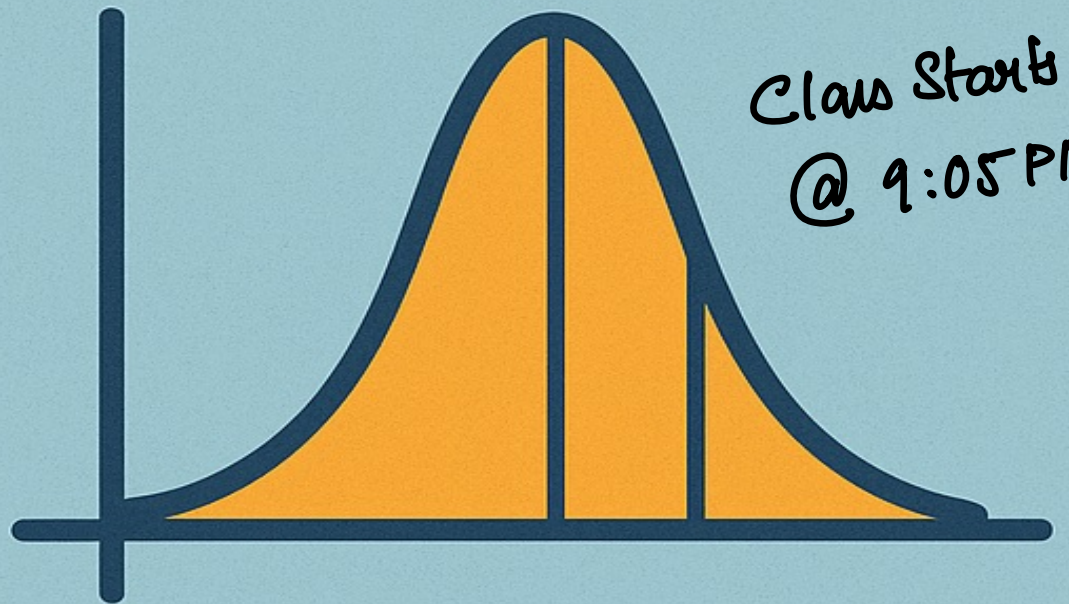
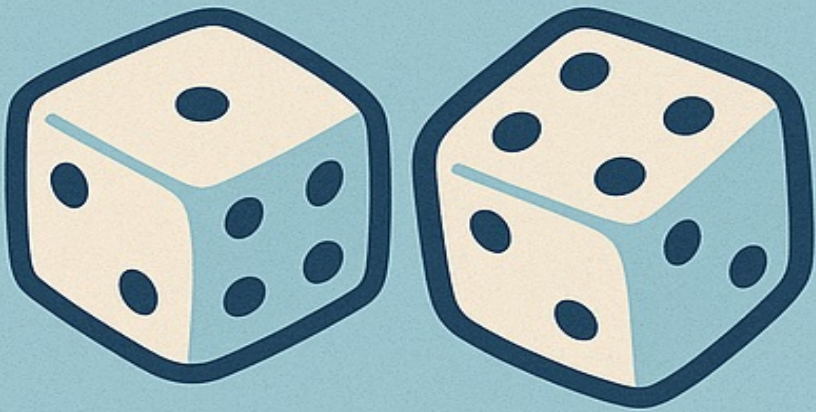






PROBABILITY AND STATISTICS



Class Starts
@ 9:05 PM

Practice Session 2

#Agenda

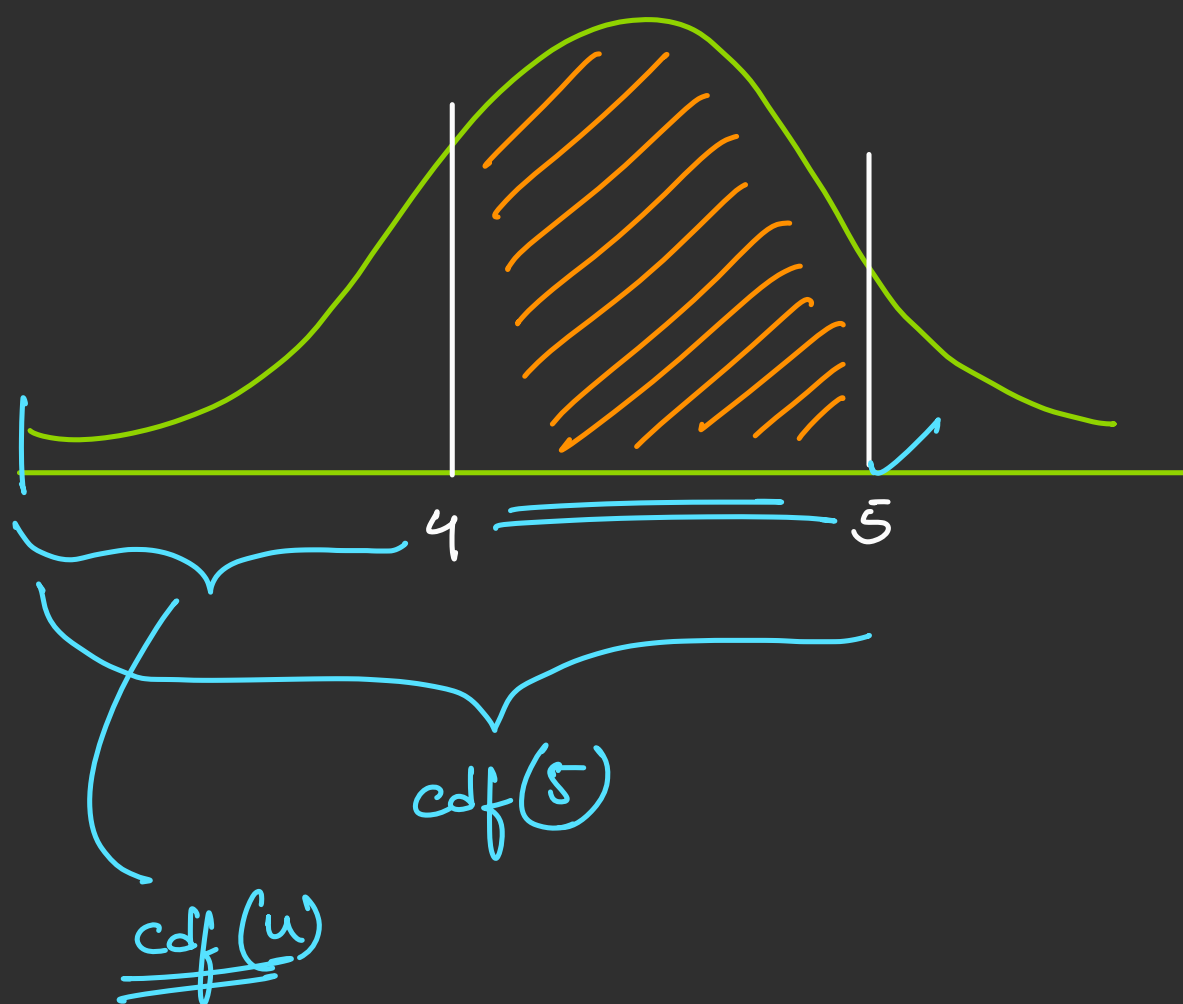
- 1) Exponential distribution
- 2) Geometric distribution
- 3) Solving problems → As many as possible
= (Homework)

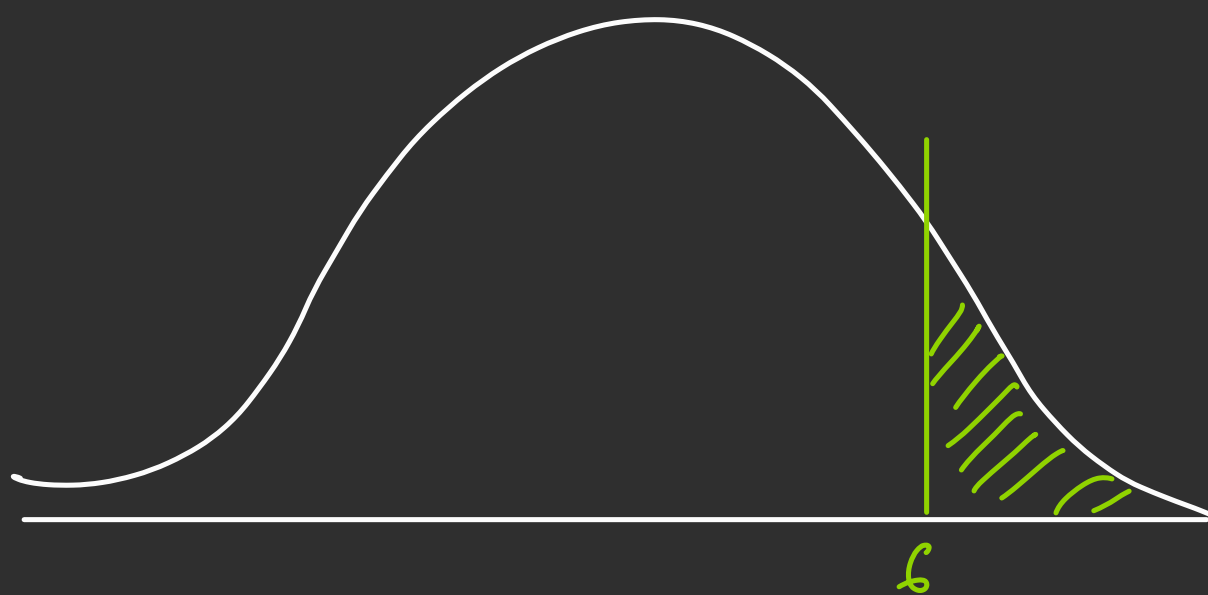
Exponential distribution

↳ Continuous dist

$$\text{scale}(\beta) = \frac{1}{\text{rate}(\lambda)}$$

$$P[4 < X < 5]$$





$$1 - \text{expon.cdf}(6)$$

Q) given I have already spent 3 mins to debug.
What is prob I need more than 9 mins?

$P(A|B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P[T > 9 | T > 3] = \frac{P[(T > 9) \cap (T > 3)]}{P(T > 3)}$$

$$\Rightarrow \frac{P[T > 9]}{P[T > 3]}$$

"Memory less property of exponential dist"

$$\boxed{\frac{P[T > 9]}{P[T > 3]} = \frac{e^{-9\lambda}}{e^{-3\lambda}}} = e^{-(9-3)\lambda} = e^{-6\lambda}$$

$$= e^{-(9-3)\lambda}$$

$$= e^{-6\lambda}$$

$$\boxed{P[T > 6] = e^{-6\lambda}}$$

$$T > 9 = e^{-\lambda x}$$

$$= e^{-9\lambda}$$

$$P[T > 9] = e^{-9\lambda}$$

$$e^{-6\lambda} = P[T > 6]$$

$$(T > 9) \rightarrow 0$$

spent 7mins
used 7min



$$P[(T > 9) \cap (T > 3)] \Rightarrow \underline{\underline{P(T > 9)}}$$

Intersection means common of both

Geometric distribution (discrete dist)

Job Search

→ You continue your search until you get one.
f
binomial

p → pass f → fail

$\{ p \} \rightarrow \text{stop}$

$\{ f p \} \rightarrow \text{stop}$

$\{ \overset{x}{f} \overset{x}{f} \overset{\checkmark}{p} \}$

f f f p

$S = \{ \boxed{p}, \boxed{f p}, f f p, f f f p \dots \}$

1 trial only you are stopping

$\boxed{f p}$ f f p

Bernoulli $\rightarrow$ single trial $\rightarrow$ discrete
Binomial $\rightarrow$ n trials $\rightarrow$ discrete
uniform $\rightarrow$ probability of events is same (both)
normal $\rightarrow$ bell curve, empirical rule, symmetrical
 $\rightarrow$ continuous
poisson $\rightarrow$ rate & time interval prob
exponential $\rightarrow$ time (until next event occurs)
 $\rightarrow$ continuous
geometric $\rightarrow$

Prob of pass (p) $\rightarrow 0.1$ ($p = 0.1$)

Prob of failure (f) $\rightarrow 1 - p$ ($f = 0.9$)

$$S = \{p, fp, ffp, fffp, \dots\}$$

$$P[X=1] = 0.1$$

$$P[X=2] = (1-p) \times p = (0.9) \times (0.1)$$

$$P[X=3] = (0.9) \times (0.9) \times (0.1) \Rightarrow (0.9)^2 \times (0.1)$$

$$P[X=4] = (0.9)^3 \times (0.1)$$

$$P[X=K] = \underbrace{(0.9)^{K-1}} \times \underbrace{(0.1)}$$

$$\Rightarrow (1-p)^{K-1} \times (p)$$

$$P[X=K] = (1-p)^{K-1} \times (p)$$

Practice Question

$$E(\underline{X^2}) = \sum \underline{x^2} \times \underline{P(X=x)}$$

Binomial dist

Exp value $E(X) \Rightarrow \underline{np}$

$$E(X^2) = \sum x^2 \times P(\underline{X=x}) \Rightarrow (0 \times 0.03) + (\underline{1 \times 0.97})$$

$\searrow (0, 1) \quad \searrow (0.03, 0.97)$

$\begin{cases} \text{CLT} \\ \text{CI} \end{cases}$

Sample Statistics

SE



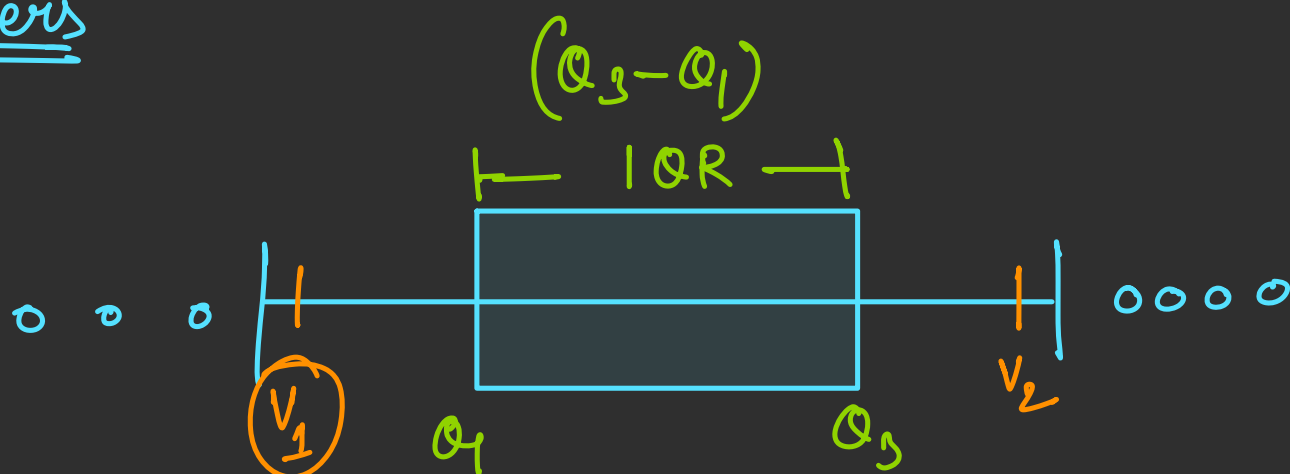
2

1

2

2
bred

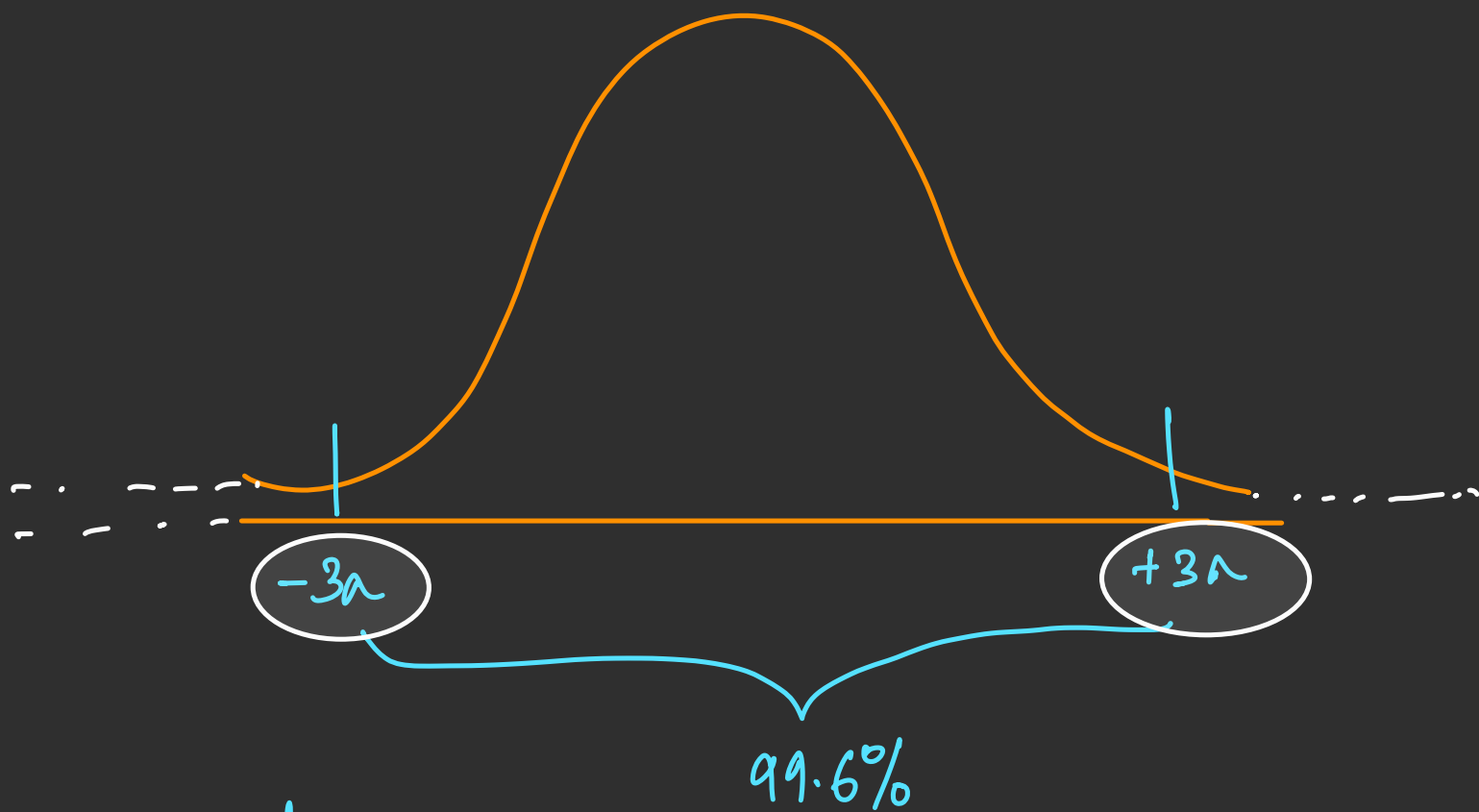
#Outliers



IQR method

$$x_1 \Rightarrow Q_1 - 1.5 \times IQR \Rightarrow \underline{\underline{value_1}}$$

$$x_2 \Rightarrow Q_3 + 1.5 \times IQR \Rightarrow \underline{\underline{value_2}}$$



<u>col 1</u>	<u>z value</u>
⋮	⋮
⋮	⋮
⋮	⋮

$$z = \frac{x - \mu}{\sigma}$$

$$\left[(z > z_1) \text{ and } (z < z_2) \right]$$

↓
✓
col1

↓
col2

index

1

|

- . - . - .

100